COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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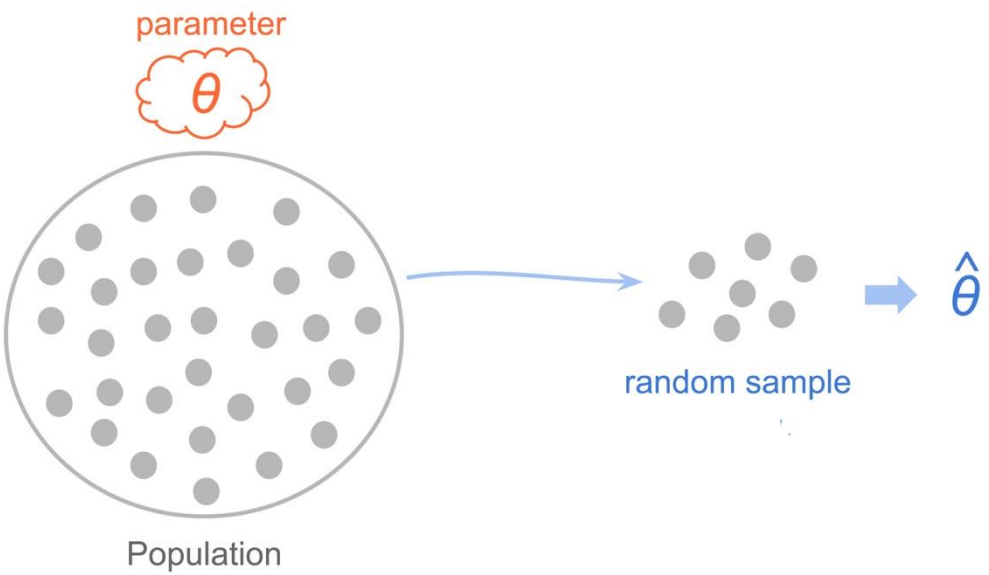
Estimation

Lecture Goals:

Define estimators
Study properties of estimators



Parametric vs. Nonparametric Models

- Our data are $X_1, \ldots, X_n$ which are iid random sample from a population with a pdf $f(x)$ or cdf $F(x)$.
- Very often the population distribution is not completely known, and $f(x)$ depends on an unknown parameter θ, which can be indicated with $f(x; \theta)$.
- When $f(x)$ depends on an unknown parameter θ, we say that we are considering a *parametric* model for our data.
- **Example**: linear regression is a parametric model.
- Non-parametric methods are statistical techniques that do not rely on specific assumptions about the underlying distribution of the population being studied.
- **Example**: K-Nearest Neighbor is a nonparametric model.
- In this course we focus on the parametric models.

Goal of Estimation

We assume we know the formula for $f(x;\ \theta)$, except for the value of θ, which is an *unknown parameter*.

Assume θ is identifiable, that is,

$$\text{if } \theta_1 \neq \theta_2, \text{ then } f(x;\ \theta_1) \neq f(x;\ \theta_2)$$

Goal: Use the observed sample values $x_1,\ x_2,\ \ldots\ ,\ x_n$ to *estimate* θ.

This means to produce a formula $\hat{\theta}_n = g(X_1, \ldots, X_n)$ such that theoretically $\hat{\theta}_n \approx \theta$

Recall: When we develop theoretical properties, we use $X_1, \ldots, X_n$ to denote the sample, emphasizing it is a collection of *random* variables. When we talk about the actual data numbers, we use $x_1, \ldots, x_n$

Example 1: Sample moments

- The sample average (aka sample mean or sample 1st moment)

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}(X_1 + \cdots + X_n)$$

is used to estimate the population mean

$$\mu = E(X) = \int x f(x) dx$$

But why? (We will recall properties of $\bar{X}$ later.)

Example 1: Sample moments

- The sample variance is defined as

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

and is used to estimate the population variance

$$\sigma^2 = \mathrm{Var}(X) = \int (x-\mu)^2 f(x)dx$$

Again, why? And why $n - 1$, not n in the denominator?

Example 2

- The order statistics of a sample $x_1, \dots, x_n$ are their values arranged in increasing order, denoted

$$x_{(1)}, x_{(2)}, \dots, x_{(n)}$$

- Often $x_{(1)}$ is called the sample minimum and $x_{(n)}$ the sample maximum.

But what are they estimating? It is not obvious, but for example, the sample median can be written as

$$\text{median}(x_i) = \begin{cases} x_{\left(\frac{n+1}{2}\right)}, & \text{if } n \text{ is odd} \\ \dfrac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}\right)+1}}{2}, & \text{if } n \text{ is even} \end{cases}$$

Definitions

Definition: Let $X_1, X_2, \ldots, X_n$ be a random sample from $f(x; \theta)$. Any formula (aka statistic)

$$\hat{\theta}_n = g(X_1, \ldots, X_n)$$

is called a *point estimator* of θ.

Definition: The *bias* of a point estimator $\hat{\theta}_n$ is defined as

$$Bias(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$$

If $E(\hat{\theta}_n) = \theta$, then $\hat{\theta}_n$ is called *unbiased*.

Example 1 (continued)

Theorem 1: Show that $E(\bar{X}) = \mu$

(prove later as exercise)

Note that this gives you a very strong result:

The sample mean is *always unbiased* estimator for the population mean (assuming it exists), regardless of what kind of population we sample from!

Exercise 1

Suppose that $X_1, \dots, X_n \sim Exp(\lambda)$

Then an estimator of λ is $\hat{\lambda} = \frac{1}{\bar{X}} = \frac{n}{\sum_{i=1}^{n} X_i}$

Show that $Bias(\hat{\lambda}) = \frac{n}{n-1}\lambda$

Hint: If $Y = \sum_{i=1}^{n} X_i$ then $Y \sim \Gamma(n, \lambda)$

Consistency

Definition: If $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$, then $\hat{\theta}_n$ is said to be a *consistent* estimator of θ.

Example 1 (continued):

Recall the weak LLN: $\bar{X} \xrightarrow{P} \mu$ as $n \to \infty$, therefore $\bar{X}$ is consistent estimator of μ. So, $\bar{X}$ is *both* unbiased and consistent!

(Now we answered the first question.)

Accuracy

- We all heard about "outliers" and how badly they affect the sample average $\bar{x}$. But what does this mean? The underlying general issue is how reliable and accurate an estimator $\hat{\theta}_n$ is.
- That is, how "close" is $\hat{\theta}_n$ to θ and under what condition it stays close.
- How do we even measure the accuracy, given the fact we don't even know the value of θ?

Standard Error

Definition: The *standard error* of an estimator $\hat{\theta}_n$ is defined as $\sqrt{\mathrm{Var}(\hat{\theta}_n)}$ and is denoted $\mathrm{SE}(\hat{\theta}_n)$.

That is,

$$\boxed{\mathrm{SE}(\hat{\theta}_n) = \sqrt{\mathrm{Var}(\hat{\theta}_n)}}$$

Mean Squared Error

Definition : The *mean squared error* (MSE) of an estimator $\hat{\theta}_n$ is defined as:

$$\boxed{\mathrm{MSE}\left(\hat{\theta}_n\right) = E\left[\left(\hat{\theta}_n - \theta\right)^2\right]}$$

Notes:

- The MSE is the risk function with respect to the squared error loss function.
- The exact formula for MSE is often too difficult to derive explicitly, so we might have to approximate it.

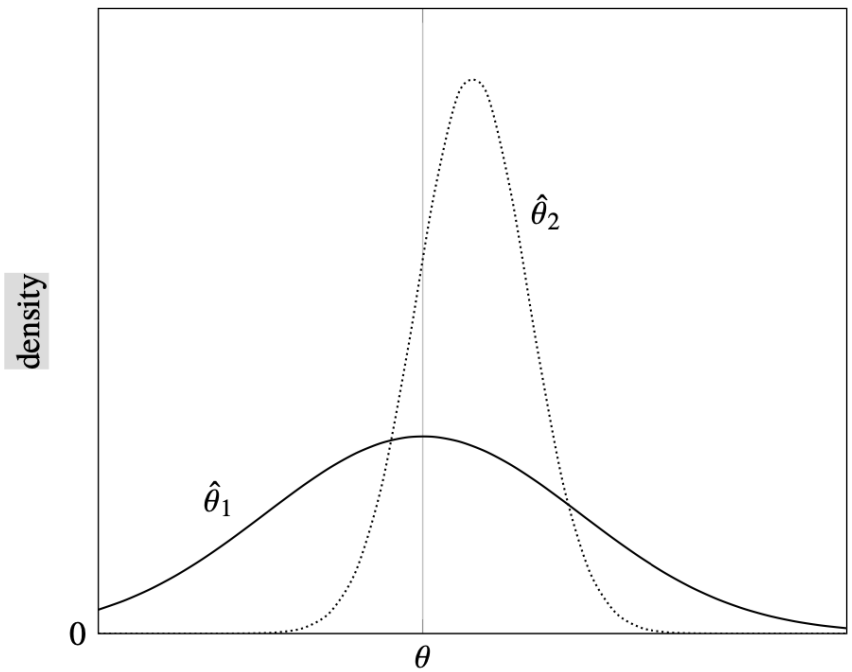
Mean Squared Error

Theorem 2:

$$\mathrm{MSE}(\hat{\theta}_n) = Var(\hat{\theta}_n) + \left[Bias(\hat{\theta}_n)\right]^2$$

Note: This result is often referred to as the **bias-variance trade off**.

- Two estimators of the same parameter θ.
- $\hat{\theta}_1$ has zero bias but high variance, so it may produce estimates very far from the true value of θ.
- By contrast, $\hat{\theta}_2$ is biased but has much lower variance, so it would typically give More accurate estimates.



Exercise 2

Recall:

- The expected value is linear: for any r.v. X and Y and constants a and b, we have $\mathrm{E}(aX + bY) = a\mathrm{E}(X) + b\mathrm{E}(Y)$
- For the variance, a similar property is

$$\mathrm{Var}\,(aX + b) = a^2\mathrm{Var}(X)$$

Use the above to show that:

$$\boxed{\mathrm{E}(\bar{X}) = \mu, \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}}$$

Note the above shows that $MSE(\bar{X}) = \frac{\sigma^2}{n}$

Examples

Example 3: *Let* $X_1, X_2, \ldots, X_n \sim$ *Bernoulli*(p), *where p is unknown* parameter. Define

$$\hat{p}_n = \frac{X}{n} = \frac{X_1 + \cdots + X_n}{n}$$

That is, $\hat{p}_n$ is the *sample proportion*.

Prove:

a) $E(\hat{p}_n) = p$

b) $\mathrm{SE}(\hat{p}_n) = \sqrt{\frac{p(1-p)}{n}}$

Proof: show on board

How to prove consistency

Theorem 3: If for all θ, $\text{Bias}(\hat{\theta}_n) \to 0$ and $\text{SE}(\hat{\theta}_n) \to 0$ as $n \to \infty$, then $\hat{\theta}_n$ is a consistent estimator of θ.

Proof:

We need to show that $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0, \forall \varepsilon > 0$

Use Chebyshev's inequality:

$$P\left[\left(\hat{\theta}_n - \theta\right)^2 \geq \varepsilon^2\right] \leq \frac{E\left[\left(\hat{\theta}_n - \theta\right)^2\right]}{\varepsilon^2}$$

$$\Rightarrow 0 \leq \lim_{n\to\infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) \leq \lim_{n\to\infty} \frac{E\left[\left(\hat{\theta}_n - \theta\right)^2\right]}{\varepsilon^2}$$

… (finish as exercise).

Example 3 (continued)

Show that $\hat{p}_n$ is a consistent estimator of p.

Proof:

$$Bias(\hat{p}_n) = 0 \to 0 \text{ as } n \to \infty$$

$$SE(\hat{p}_n) = \frac{\sqrt{p(1-p)}}{\sqrt{n}} \to 0 \text{ as } n \to \infty$$

Therefore, $\hat{p}_n$ is a consistent estimator of p.

Mean Squared Error Example

Example 3 (continued):

c) Find $\mathrm{MSE}(\hat{p}_n)$

Solution:

$$\mathrm{MSE}(\hat{p}_n) = Var(\hat{p}_n) + [Bias(\hat{p}_n)]^2$$

$$= \frac{p(1-p)}{n} + 0^2 = \frac{p(1-p)}{n}$$

Exercise 3

Example 3 (continued)

Finish as exercise:

b) Consider an alternative estimator (sometimes called Wilson's adjustment)

$$\tilde{p}_n = \frac{X + 2}{n + 4}$$

Find its bias and variance.

Mean Squared Error

b) Consider an alternative estimator

$$\tilde{p}_n = \frac{X + 2}{n + 4}$$

Find its bias and variance.

After completing the exercise, you will find out that:

- $Bias(\tilde{p}_n) \neq 0$
- $Var(\tilde{p}_n) < Var(\hat{p}_n)$

Exercise 4*

Recall that $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$

a) Show that

$$MSE(S^2) = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right)$$

b) Assume $X_1, \ldots, X_n \sim \mathrm{N}(\theta, \theta)$ and consider two estimators of θ: $\bar{X}$ and S. Compare the MSE's.

Example 4:

Suppose the following observations are from a *normal* distribution with unknown μ and σ:

0.83, 0.88, 0.88, 1.04, 1.09, 1.12, 1.29, 1.31, 1.48, 1.49, 1.59, 1.62, 1.59, 1.62, 1.65, 1.71, 1.76, 1.83

$\bar{x} = 1.377$, $s = 0.3287$

a) Estimate μ;

b) Estimate the median;

c) Estimate the 90th percentile.

Example: Distribution of the Max

Let $X_1, X_2, \ldots, X_n$ denote a random sample of size n from a population with pdf $f(x)$ and cdf $F(x)$

Define $M = \max(X_i) = X_{(n)}$

Task: determine the pdf $g(x)$ of M.

Repeat this as exercise for $m = \min(X_i) = X_{(1)}$

Solution:

Find the cdf, $G(x)$ of M:

$$G(x) = P(M \leq x) = P(\max\{X_1, \dots, X_n\} \leq x)$$
$$= P(X_1 \leq x, \dots, X_n \leq x)$$
$$= P(X_1 \leq x) \times \cdots \times P(X_n \leq x)$$
$$= F^n(x)$$

Now, differentiate to find the pdf:

$$g(x) = G'(x) = nF^{n-1}(x)F'(x)$$

Therefore,

$$g(x) = nf(x)F^{n-1}(x)$$

In general, the order statistics of a random sample $X_1, X_2, \ldots, X_n$ are the sample values placed in ascending order and denoted $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$.

The distribution of each of them can be derived using similar arguments as for min and max.

For example,

$$\{X_{(k)} \leq x\} = \bigcup_{j=k}^{n} \{At\ least\ j\ Xs < x, n - j\ Xs > x\}$$

Exercise 5

Derive the pdf of the minimum m of uniformly distributed data between 0 and θ.

Answer:

$$g(t) = \frac{n}{\theta}\left(1 - \frac{t}{\theta}\right)^{n-1}, \qquad 0 \le t \le \theta$$

Exercise 6

Let $X_1, \ldots, X_n \sim \mathrm{U}(0, \theta)$, where $n = 6$. That is,

$$f(x) = \frac{1}{\theta}, 0 \leq x \leq \theta$$

Define $\hat{\theta} = \max\{X_1, \ldots, X_n\}$ to be an estimator of θ.

Calculate the probability that $\hat{\theta}$ falls within 0.2 of θ given that the parameter θ true value is 3.

Answer: 0.338971

Example 5

Let $X_1, \ldots, X_n \sim \mathrm{U}(0, \theta)$.
That is,

$$f(x;\theta) = \frac{1}{\theta}, 0 \leq x \leq \theta$$

Define $\hat{\theta} = \max\{X_1, \ldots, X_n\}$ to be an estimator of θ.

a) Is $\hat{\theta}$ unbiased?
b) Is $\hat{\theta}$ consistent?

Unbiased estimators

Example 5: Let X_1, X_2, ... , X_n be a random sample from the $U(0, \theta)$ distribution, where θ is unknown. Define the estimator

$$\hat{\theta} = \max\{X_1, \dots, X_n\}$$

Answers to Example 5:

a) $\text{Bias}(\hat{\theta}) = -\frac{\theta}{n+1}$

b) $\text{Var}(\hat{\theta}) = \frac{n\theta^2}{(n+2)(n+1)^2}$

Example 5

Let $X_1, \ldots, X_n \sim \mathrm{U}(0, \theta)$. That is, $f(x) = \frac{1}{\theta}, 0 \leq x \leq \theta$.

Define $\hat{\theta} = \max\{X_1, \ldots, X_n\}$ to be an estimator of θ.

a) Is $\hat{\theta}$ unbiased?

Let $g(x)$ be the pdf of $\hat{\theta}$. Then $g(x) = nf(x)F^{n-1}(x)$

$$\Rightarrow E(\hat{\theta}) = \int_0^\theta xg(x)dx = \int_0^\theta xn\frac{1}{\theta}\left(\frac{x}{\theta}\right)^{n-1} dx = \frac{n\theta}{n+1}$$

$$\Rightarrow \quad \mathrm{Bias}(\hat{\theta}) = \frac{n\theta}{n+1} - \theta = -\frac{\theta}{n+1}$$